

Reactions

Reactions are part of the natural course of leprosy infection and can be triggered or modified by treatment. There are three main types: **type I leprosy reaction (T1R)**, also known as **reversal reaction (RR)**; **type II leprosy reaction (T2R)**, also called **erythema nodosum leprosum (ENL)**; and **Lucio's phenomenon**, which occurs specifically in multibacillary patients, particularly in Mexico.

Type I Leprosy Reaction (T1R)

T1R is a **cell-mediated immunity (CMI) reaction**, classified as a **type IV Gell and Coombs hypersensitivity reaction** directed against *M. leprae* antigenic determinants.

Clinical features: - Increased inflammation of existing skin lesions: they become more **erythematous, raised, and enlarged**, and may **ulcerate**. - **Nerve involvement:** nerves may become **inflamed, enlarged, and tender**, leading to: - Loss of strength - Reduced sensation - Decreased sweating - **Acro-oedema** (swelling of hands and feet) - Involvement of joints and liver may also occur.

Type II Leprosy Reaction (T2R)

T2R is primarily an **immune complex-mediated (antigen-antibody) reaction**, affecting tissues such as skin and nerves. However, **cell-mediated immunity and T-cell regulation** also play a role.

Clinical features: - Characteristic **red, painful, tender nodules** in the skin, which may: - Ulcerate - Resemble erythema multiforme - Develop into bullous lesions - Systemic involvement (multi-organ disease), including: - Neuritis - Lymphadenitis - Orchitis - Iridocyclitis - Arthritis - Hepatitis - General symptoms: **fever, leucocytosis**, and occasionally **lymphocytes in urine**.

Treatment of Type I Reaction (T1R)

The primary treatment is **corticosteroids**: - **Starting dose**: 30–40 mg prednisone daily - **Tapering**: guided by clinical assessments such as: - Graded sensory testing (e.g., Semmes-Weinstein monofilaments) - Voluntary muscle testing - **Duration**: 4–12 months - **Minimum effective dose**: 15–20 mg prednisone - **Adequate immune suppression** (>0.25 mg/kg/day) should be maintained for: - At least **3 months** in borderline tuberculoid (BT) leprosy - **6–12 months or longer** in borderline lepromatous (BL) leprosy

Recurrence

After treatment, leprosy may recur due to: - Undertreatment - Drug resistance - Persister organisms - New infection

Recurrences are generally **sensitive to the original multidrug therapy (MDT)**. However, resistance has been documented to: - **Dapsone** - **Rifampicin** - **Ofloxacin**

Resistance to **clofazimine** has not been convincingly proven.

A **PCR test** is available for detection and monitoring.

WHO Recommendations for Resistant M. leprae

For resistant cases, WHO recommends modifying MDT based on resistance patterns, using a combination of three drugs from the following: - Ofloxacin - Clarithromycin - Minocycline - Clofazimine